



University
Mohammed VI
Polytechnic



Deliverable : Logical design Lab2

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	Team N°1
Member 1	Nizar El HAZZITI
Member 2	Mohamed JBARI
Member 3	Riyad HABIBI
Member 4	Chaimae IAA
Repository Link	https://github.com/...

1 Introduction

This lab focuses on the logical design phase of the RLMS project, which consists of translating the ER diagram built in Lab 1 into actual database tables with rows, columns and primary keys while preserving the semantics captured in the conceptual model.

2 Requirements

This deliverable translates the RLMS ER diagram into a relational schema, mapping all entity sets, relationships (including weak entities, ISA hierarchies, recursive and ternary relationships, and aggregation) into properly keyed tables, specifying primary keys, foreign keys. It also identifies any constraints not expressible via keys alone, and delivers the full .sql schema plus a short report on the team, challenges, and design choices.

3 Team roles

Nizar: Modelizing relational table for entities Person and Laboratories.

4 Key challenges

- For Key constraints we won't need an individual table for such a relationship, therefore we add one attribute to the entity, otherwise we create another table for the relationship.
- For M:1 relationships (like monitors) no need to create a new table, we just add an attribute technically to the laboStock relationship, since each stock has only one monitor.
- "Supplier" is a ternary relationship with attribute "unitPrice", the attribute then depends on the combination of the three entities (lab, supplier, consumable), so we can't split it into labSupp and consumSupp.

5 Discussion

Lessons learned:

- We don't split a ternary relationship into two tables ! It stays one table with the 3 Foreign Keys.
- When merging a relationship into another table, we NEVER change the primary key (like monitors).
- A relationship based on top of an aggregated relationship (like stocks) must reference that aggregate's one table NOT the original entities it was built from.

6 Conclusion and thoughts for next lab

A simple .sql file where we implemented the relational model. The only thing to address better is the consistency of code style, because each member has his style (example:

the use of camel or snake case, tabulations ...), we should find a good style and all of us use it.